gearing for torque reduction is then usually provided by spur gears, although a further change of direction via a bevel gear may sometimes be needed.

Hypocycloidal gear-train manual actuators have been especially developed for the operation of centred-disc butterfly valves mainly used in the heating, ventilation and air conditioning industry (HVAC).

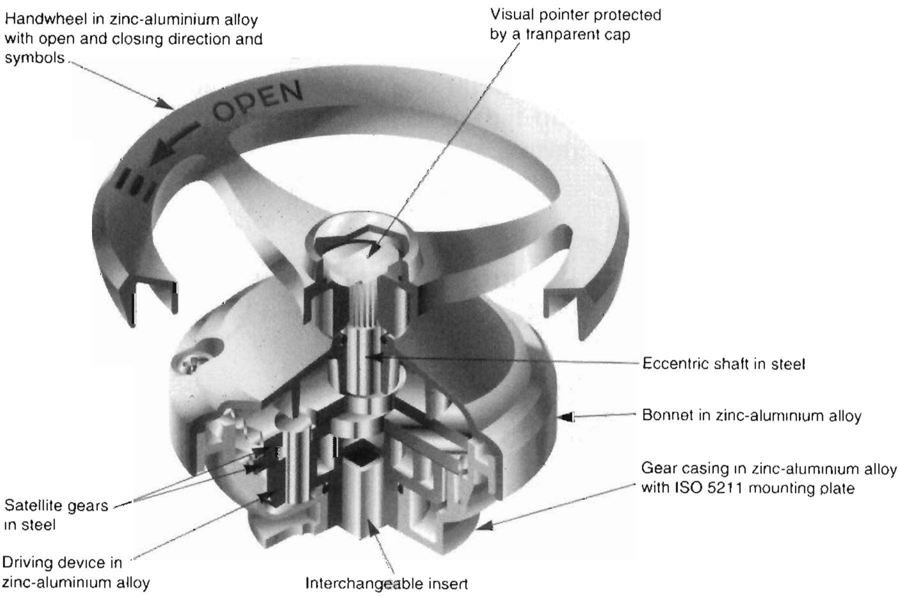
Other manual actuators with worm wheel and screw kinematics are used with $^1/_4$-turn valves including centred or double-eccentric disc butterfly valves, ball valves, etc. They are designed to deliver a constant output torque.

Signalling

A problem with manually-operated valves—especially in plants with centralised control—is the absence, usually, of any standard means of signalling the valve status, or of confirming that a required operation has been carried out. If this is necessary, special provision has to be made. Often this is done by attaching some form of external limit-switching mechanism, using switchboxes that meet environmental and safety requirements (see the section on Limit switches below).

Handwheel drives for powered actuators

All the aforementioned points apply to auxiliary manual drives for power actuators. However, the fact that the drive is auxiliary only and not the main



Hypocycloidal gear-train kinematics.

means of operation may make design compromises in the interests of economy more acceptable than they would be if manual operation were the regular and only means.

Cylinder actuators

This type has an actuator using a piston moving inside a cylinder by pneumatic or hydraulic pressure. It can be single-acting, i.e. equipped with a return spring, or double-acting, using air or oil pressure for movement in both directions.

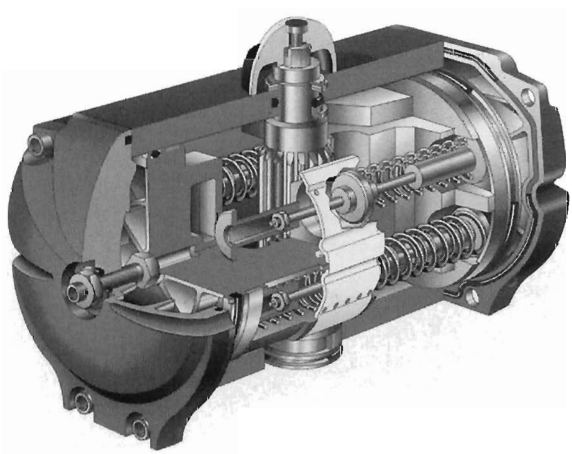
The piston and cylinder can be practically any length or diameter and readily converts pneumatic or hydraulic pressure into linear force. This can be further converted to part-turn operation by rack-and-pinion or linkage, i.e. scotch yoke.

Pneumatic actuators for industrial valves are predominantly applied in continuous processes. While one actuator may pass through hundreds of cycles, 24 hours a day, another may open and close just once a month. Pneumatic actuators can be applied to ball valves, plug valves and butterfly valves.

Vane actuators

A vane actuator is a pneumatically or hydraulically operated actuator used with rotary valves. Typically, the actuator has a paddle-like vane in a sector-shaped pressure casing giving a 90° rotary motion to the vane shaft connected to the valve stem.

Vane actuators can be double-acting, i.e. operated in both directions by air or gas pressure, or single-acting with a return spring.



Pneumatic spring-return actuator.

Electric solenoid

The electric solenoid tends to be limited to very small powers, i.e. to pilot duty rather than actuation duties for other than the smallest valves.

Diaphragm actuators

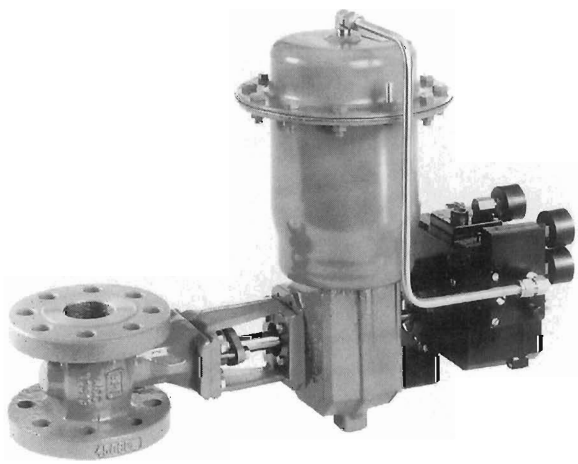
This can be described as a pneumatically operated actuator where the air chamber is sealed by a flexible diaphragm, with most of its flat area supported by a plate at the end of an actuator stem. Variable air pressure flexes the diaphragm and positions the stem with the assistance of a return spring. Spring-diaphragm actuators are designed for both proportional and on–off control of rotary valves. They may be operated by air, gas, water, oil or other supply media compatible with the diaphragm and its case.

Other types of spring-diaphragm pneumatic linear actuators are designed for operation with linear control valves. The action can be direct-acting, where the stem extends with increased air pressure, or reverse-acting, where the stem retracts with increased air pressure.

Electric-motor actuators

Stringent demands are made of electric actuators for industrial valves. Extreme temperature fluctuations and aggressive media that affect the actuator's resistance to chemicals are just two of them.

Microcomputers are being increasingly used to achieve more accurate and finer controls and adjustments. They are installed not only in large facilities but also in individual equipment. The trend will intensify with price reductions



Spring-diaphragm actuator.

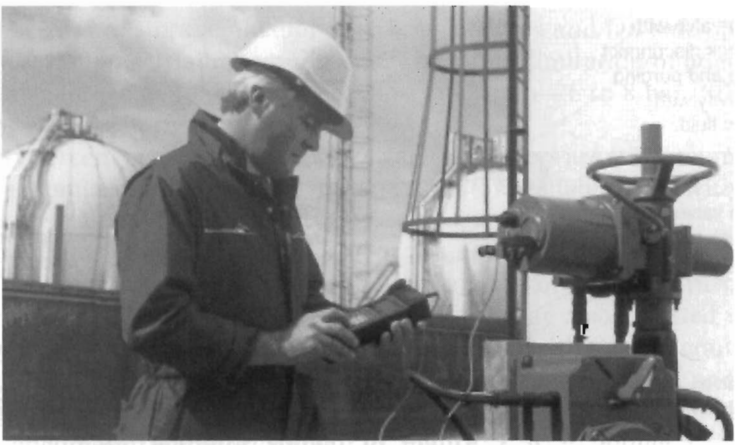
in microcomputers and with the desire to trace system hazards quickly. These factors have caused an acute need for automatic controls for entire piping systems.

Generally, electric-motor actuators are designed for use on ball valves, gate valves, butterfly valves, plug valves and any mechanical equipment calling for 90° rotational control, including dampers and ventilation grids, etc.

The development of the smart valve accessible by a digital communications link means that commands can initiate a stroke check of the valve, recording pressure versus valve travel as the valve is stroked. With an additional sensor for stem and shaft position, the data can then be used to evaluate the condition of the actuator and accessories under various parameters based on an investigation of the valve's stored history. Digital communications to the valve can measure input signal, pneumatic pressure and valve travel, comparing the data with stored expected value's and recommending corrective action. The electric motor will become a serious challenge to pneumatic power when its inertia matches that of a piston or a diaphragm and when gears have zero backlash. It must also rid itself of thermal overloads, limit switches, cams, heaters and thermostats, duty-cycle limitations and explosion-proof housings.

The trend, though, is definitely towards electronics and micro-electronics and for the future there is the possibility of an actuator that emulates a biological muscle—a fast, powerful mechanism that uses stored chemical energy and is controlled by weak electrical pulses. This device might consist of polymer strands that contract on signal and take their energy from a chemical bath that is recharged electrically, as needed by a continuously connected power source.

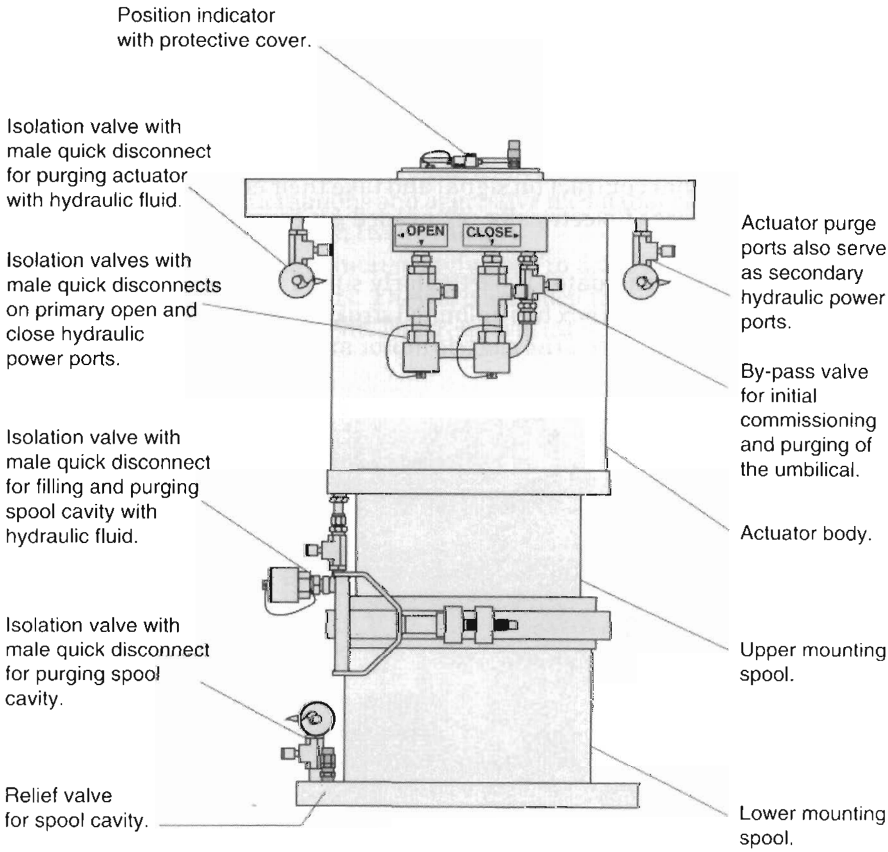
The electric-motor actuator is particularly suited where the stroke is long, because a motor with gearbox has unlimited stroke. However, there are different mechanical virtues between the electric motor and the piston and cylinder. As



Intelligent communication and control.

mentioned earlier, the piston can be kept continuously pressurised, whereas the electric motor cannot; a piston can maintain its position, but the drive of a motor must be mechanically self-locking in order to maintain its position when switched off. The motor does, however, have the virtue of kinetic energy—the ability to take a 'running jump'. Kinetic energy, however, places mechanical constraints on the use of motor actuators as the energy must be controlled and not misapplied.

In certain circumstances, electric-motor actuators are not very efficient. For example, when driving through a worm gear and nut-and-screw, e.g. on a gate valve, up to 90% of the motive power is trying to wear itself out and only 10% is useful. This is acceptable for intermittent valve operation, but it is one of the reasons why geared motor actuators are not well suited to continuous modulating or positioning control via stem nuts or self-locking gears. Multi-turn electric modulating actuators with linear output drives may be a better solution.



Rotary-vane subsea actuator.

Choice of energy system

Clearly the selection of the energy system for a particular valve-operating duty is not something that can be made in isolation. Overall design considerations, safety requirements, availability of supplies and total installed initial cost and subsequent maintenance costs all need to be considered.

No single type of control valve actuator is best for all applications. Demands for power, speed, stiffness and precision vary and cost considerations are always present. For some applications, there is no actuator that performs adequately.

Sometimes practical valve or environmental needs dictate the use of a specific type of actuator, in which case the energy source is predetermined. Sometimes the non-availability of an energy source—or the prohibitive cost of providing it—will rule out certain actuator options. What follows, therefore, is only a brief resumé of the basic systems available, and the practical and economic pros and cons of each for various duties. This must, of course, be related to the foregoing section on actuator types and their performance capabilities.

Advice is readily available from most actuator manufacturers on the choice, sizing, adaption, installation and commissioning of valve-control systems. For the unwary or inexperienced specifier, there may be some risk of bias in such advice from manufacturers of particular types of actuator. This is rare among leading suppliers whose own reputation must stand or fall on the soundness of advice in terms of performance and reliability.

Pneumatic systems

If the operation is to be within the confines of a plant where compressed air is available, the cheapest method of valve actuation is a pneumatic piston and cylinder. Pneumatic operation over a distance is, however, limited entirely by the high cost of making adequate compressed air available over long distances, ease of storage of compressed air facilities and fail-safe operation under electric-power failure conditions. The major limitation of pneumatics is available force due to practical pressure limitation—6 to 8 bar (80 to 100 lbf/in^2) being the normal maximum.

Ninety percent of modulating control valve actuators are still pneumatic.

There are many variations of pneumatic actuator available today with output torques up to 12,000 Nm (9000 ft.lbs).

The actuator shown in Figure 1 is of the double-acting rack-and-pinion type and is particularly suitable for the automation of a quarter-turn valves (butterfly snd ball valves). This type of actuator may also be fitted with an electronic integrated instrumentation unit to ensure the direct control and supervisory functions encountered in all modern processes and, more particularly, in communication by fieldbus.

The pneumatic actuator shown in Figure 2 is a double-acting type incorporating scotch yoke drive.

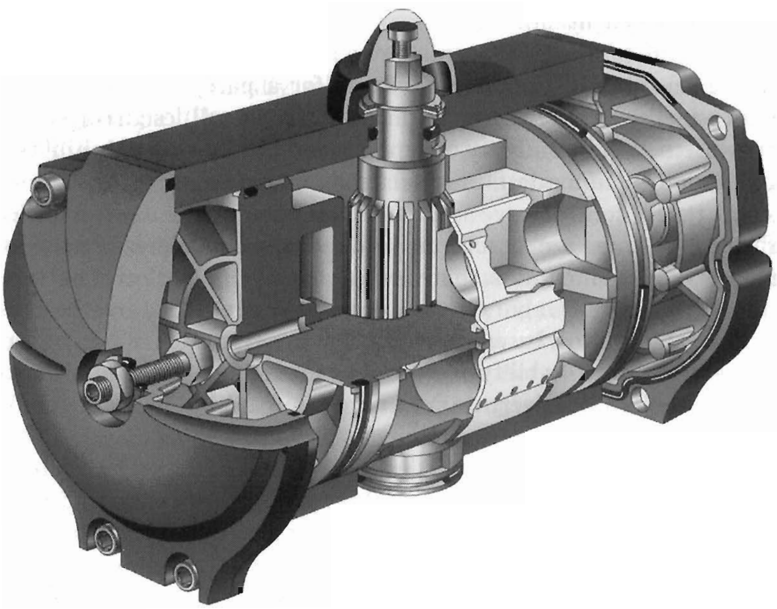


Figure 1. Double-acting rack-and-pinion pneumatic actuator.

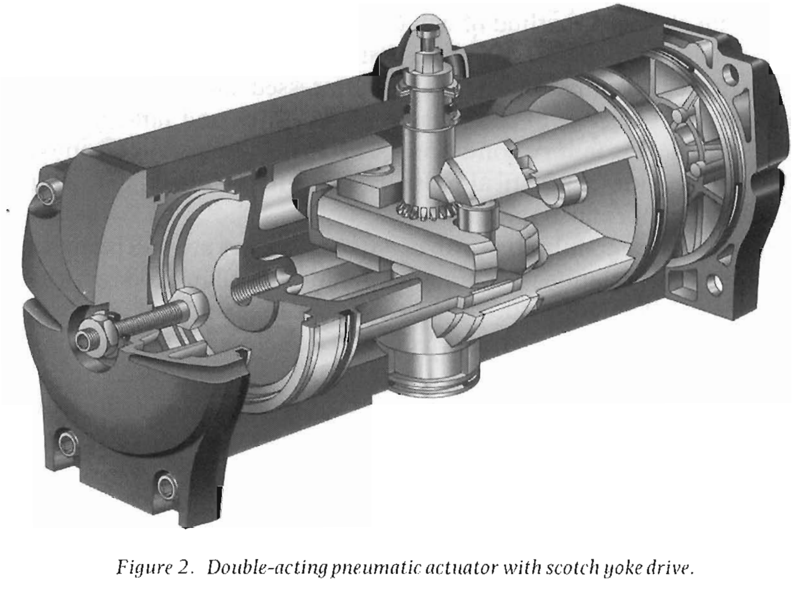


Figure 2. Double-acting pneumatic actuator with scotch yoke drive.

This actuator develops a variable torque and is well suited for the operation of larger size quarter-turn valves (butterfly and ball valves) when a significant torque is needed near the closed position or near the open position. A typical standard type of double-acting and spring-return piston-type actuator is shown in Figure 3.

This style of pneumatic actuator also employs a self-contained spring cartridge to protect against failure in either the open or closed position. Actuators of this type offer an extremely long cycle life and are well suited to operate almost any rotary valve in both modulating control and on–off service. Most actuators of this type are located in fully-closed housings, sealed against humidity and dust, and should not need regular maintenance. Another versatile unit is the positioner–actuator which consists of a double or single actuator, a pneumatic or I/P positioner and conventional or inductive limit switches. This is mainly used with small, segmented ball valves.

Normal operation of pneumatic actuators is generally accomplished by pressurising the appropriate supply ports by means of an air-control valve. Most solenoid and control valves perform better on lubricated air which may be added with an air line lubricator. Clean, dry air extends the life of pneumatic actuators and, if this is not available, an in-line filter should be used. Before hook-up, air lines should be purged to remove scale and other particles which could damage the control valve, positioner and actuator seals.

Figure 3. Standard-type double-acting and spring-return pneumatic actuator.

Vane systems

Rotary-vane valve actuators are used for quarter-turn valves in critical applications, especially on natural gas pipelines where the actuators are typically powered by natural gas using gas over oil tanks. Other applications include:

- quarter-turn valve control on crude-oil products pipelines with the actuators powered hydraulically or by nitrogen-storage vessels
- high-vibration applications including slurry-pipeline valves, pumping station valves and compressor-station valves
- offshore platform applications
- cryogenic or extremely low-temperature applications
- subsea-valve control
- high-speed applications with stroking times as fast as 250 ms

A typical rotary-vane actuator is shown in Figure 4.

The general principle of operation of this type of actuator is that opposite chambers in the actuator are corrected by pressure-equalising passages in the upper and lower heads. In this manner, the actuator produces a balanced torque as hydraulic force simultaneously pushes both of the rotor valves away from the stationary shoes.

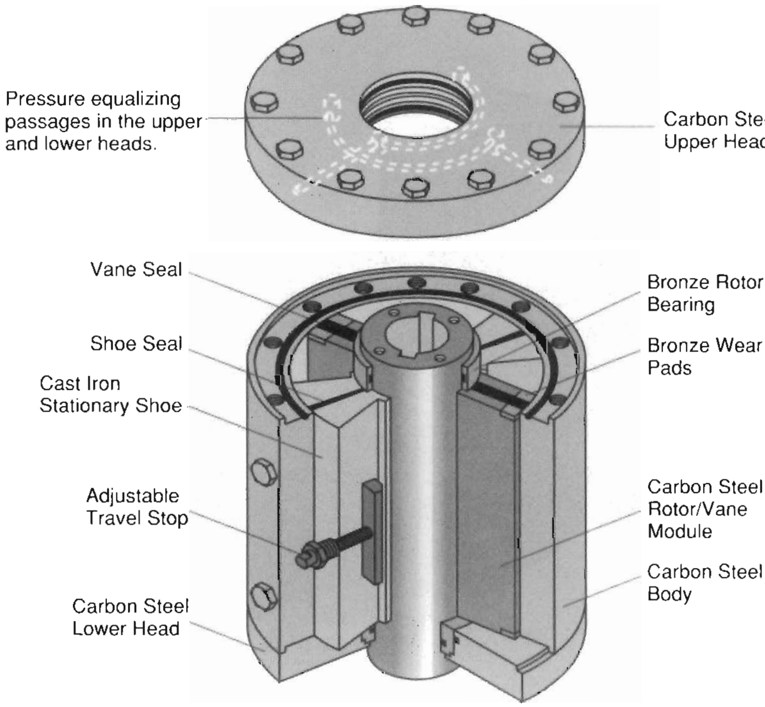
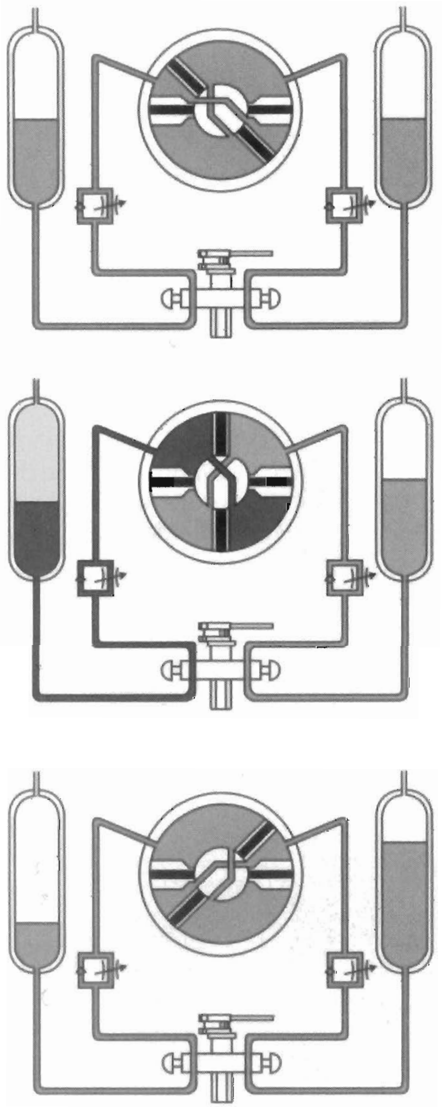


Figure 4. Rotary-vane actuator: schematic.

Torque output of the rotary-valve actuator remains constant throughout the full rotation of the valve. Constant torque output is an especially important feature in high-flow applications, plug-valve applications and for valves which have rotating seats. Constant torque output insures that the specified safety factor will not diminish at various positions during the valve stroke.

The rotary-vane actuator has often been described as the complete actuator for high-pressure duties. Regulators, pressure-reducing valves or relief valves



SEQUENCE 1

The actuator may be powered by a hydraulic power unit, stored gas pressure or by natural gas pressure from a pipeline. In this illustration, the actuator is fitted with gas hydraulic tanks and is powered by gas pressure. In the first sequence, the actuator is in the open position. There is no pressure in the actuator or tanks.

SEQUENCE 2

The actuator control system is used to admit high pressure gas into the closing gas hydraulic tank. The pressurized gas in the tank forces hydraulic fluid into the actuators closing port. Pressure equalizing passages allow both closing quadrants to be pressurized simultaneously providing balanced torque as the vanes push away from the stationary shoes. The actuator is rotating clockwise.

SEQUENCE 3

When the actuator reaches the fully closed position, the control system will allow all remaining pressure in the tank to vent to atmosphere, thus neutralizing the pressure in the tank and actuator.

High Pressure Gas

Pressurized Hydraulic Fluid

Non-Pressurized Hydraulic Fluid

Figure 5. Rotary-valve actuator: operating sequence.

are generally not required in the power supply circuit. The operating sequence of a typical rotary-valve actuator is described in Figure 5.

Hydraulic systems

Hydraulic systems are the natural choice if an exceptional duty requires a large amount of stored power (Figure 6). The forces available, and the speed

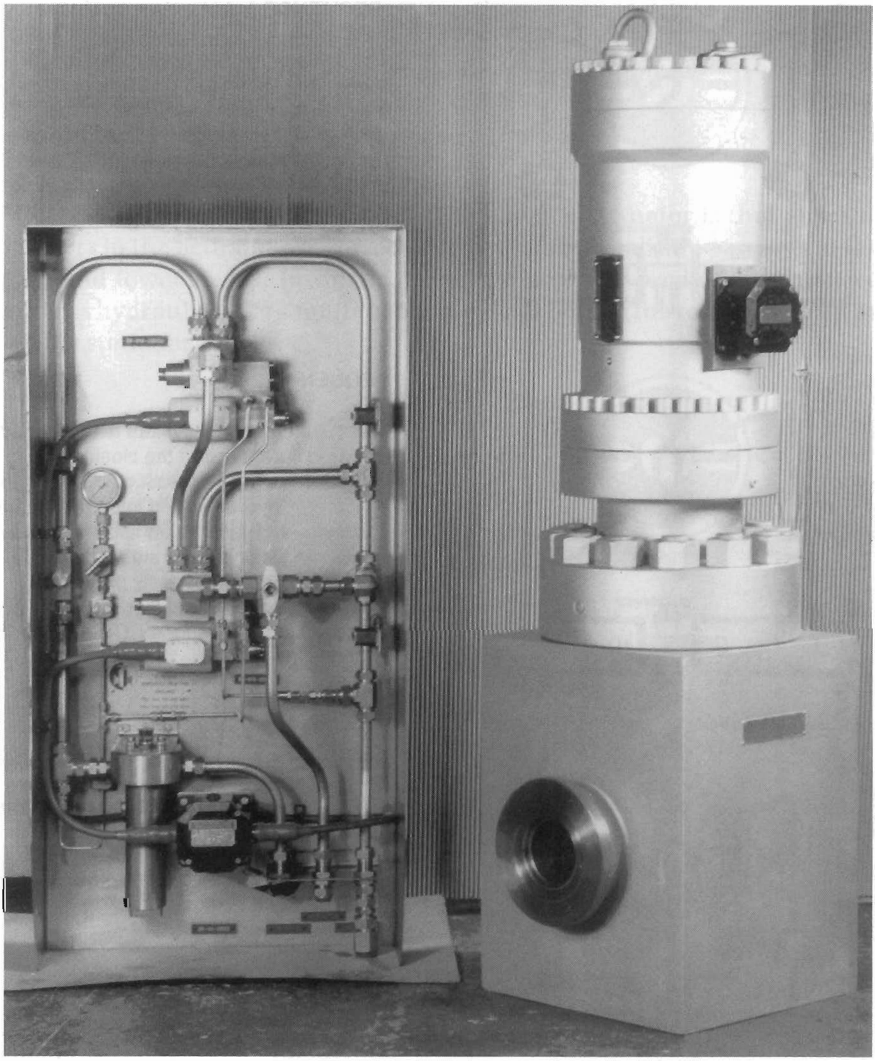


Figure 6. 15,000 lb/in² conduit-gate valve with hydraulic spring-return actuator and emergency shut-down control panel.

and control of the hydraulic actuator are virtually unlimited, i.e. the only method of operating a large valve in seconds is to pump up a hydraulic accumulator first. Unless these particular virtues are needed, the system is uneconomical and, for this reason, is not commonplace.

Furthermore, the hydraulic system is very expensive for operation over long distances. A pneumatically-operated valve is vented to atmosphere but, with the hydraulic system, the hydraulic fluid has to be returned through a line of greater suction to avoid pressure drop. For some special situations, such as tanker-cargo valves, however, hydraulics provide the only acceptable means of centralised control. Hydraulic systems are still used for fail-safe and standby duties.

Combined pneumatic/hydraulic systems (air–oil cylinder)

There are some specialised applications where the combination of pneumatics and hydraulics can meet particular operational, fail-safe and other needs. Usually, the principle involved is to use a plant's existing compressed-air supply to drive a hydraulic pump which, in turn, is used as the regular valve-actuator energy source, but which also tops up an accumulator for standby emergency use. Such systems can be devised to cater for virtually any supply-failure conditions. A typical application is for self-powered actuators in gas pipelines from which the live gas can pressurise a hydraulic accumulator system for emergency isolating valves. Live gas is also used to operate the actuator directly, as previously described.

Hydraulic actuators with output torque valves to 2000 Nm.

Electric systems

There are two main factors to be considered here: an electric motor as the power source, and electricity for the control system. Undoubtedly, full electric operation and control offers the greatest flexibility, and best suitability, to centralised automatic control. Remote indication by electrical position feedback and provision for stand-by manual operation are inherent if the system is totally electric.

Electric control

Much of the development in industry today is in the direction of centralised control, which in turn involves remote electrical control of valves, so a close look at electric valve actuators and their control systems is justified.

Electric valve actuation continues to evolve with astonishing speed. Among the leaders in the field, this evolution has resulted in very advanced, sophisticated designs suitable for instant adaptation to practically any kind of valve, in any environment, in any part of the world.

The weakness of electrical equipment is that electricity and water do not mix. However, electric actuators are required to operate in a wide variety of environments: high temperatures, steamy atmospheres, adverse weather conditions, mud, sand, flooding and so on. Thus the primary problem for the designer is not the mechanical duty—that is negligible—but to keep the environment out.

For example, a plant operating 5 days a week, 50 weeks a year, with a motorised valve of 1 minute stroke time, opening in the morning and closing in the evening, would demand a total actuator working life of only 7 days in 20 years. Thus, proper environmental sealing to enable an actuator to do nothing with complete reliability is paramount. A minimum requirement is that it should be watertight and flood-proof.

As a further safeguard to exclude the environment, the terminal area—which has to be exposed for wiring on site—should be separately sealed from the rest of the equipment, to prevent the ingress of dirt or water during installation, the most vulnerable period in an actuator's life.

Single-phase, watertight electric actuators are a simple and cost effective way of controlling small quarter-turn valves and dampers. They are suitable for use in many areas where an IP68 (NEMA 6) enclosure is required.

The example shown in Figure 7 is suitable for simple open–close duties where on–off control is required. This is achieved without the need for reversing contactors. The structural components of a typical standard electric valve actuator for use with cast-iron and stainless-steel ball valves and butterfly valves, including also motor-operated control valves and rotary control valves, are shown in Figure 8.

One example of how solid-state technology is being applied to electric motor controlled actuators is shown in Figure 9.

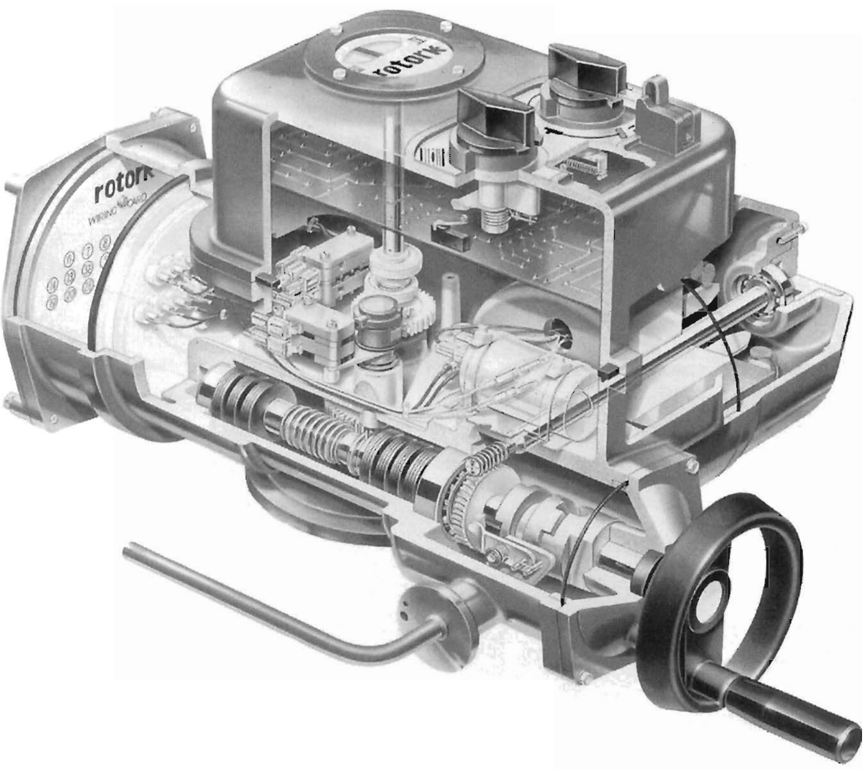


Figure 7. Single-phase electric actuator.

The actuator consists of a motor controlled by an integral solid-state control assembly driving through two stages of worm and wheel gearing to a quarter-turn output assembly giving clockwise-to-close output.

The solid-state assembly consists of two elements, the transformer rectifier providing DC power via thyristors to the motor and the CMOS gate array which controls and monitors the actuator functions and interfaces with remote controls. The logic circuits are protected from high-voltage transients which may appear at the actuator terminal by opto isolators. A schematic (Figure 10) shows the circuitry.

Electrically operated/electronically controlled

Intelligent, non-intrusive, three-phase electric valve actuators incorporate the latest electronic techniques and combine them with tried and tested motor and gearing technology (Figure 11).

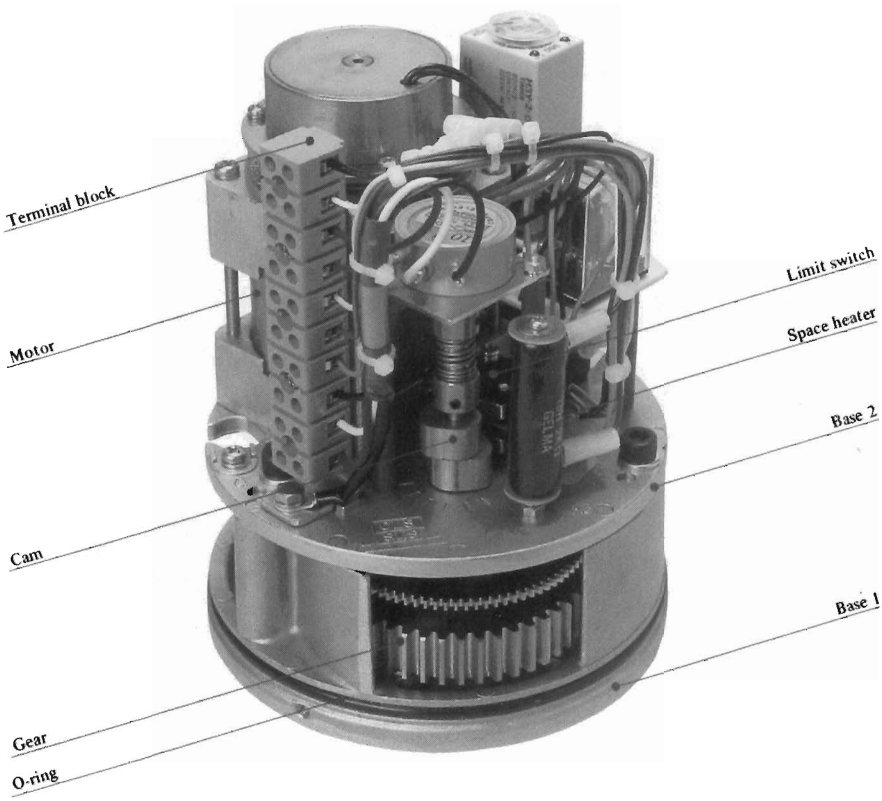


Figure 8. Electric valve actuator: structural components.

Figure 9. Electric quarter-turn actuator for fully-modulating duty.

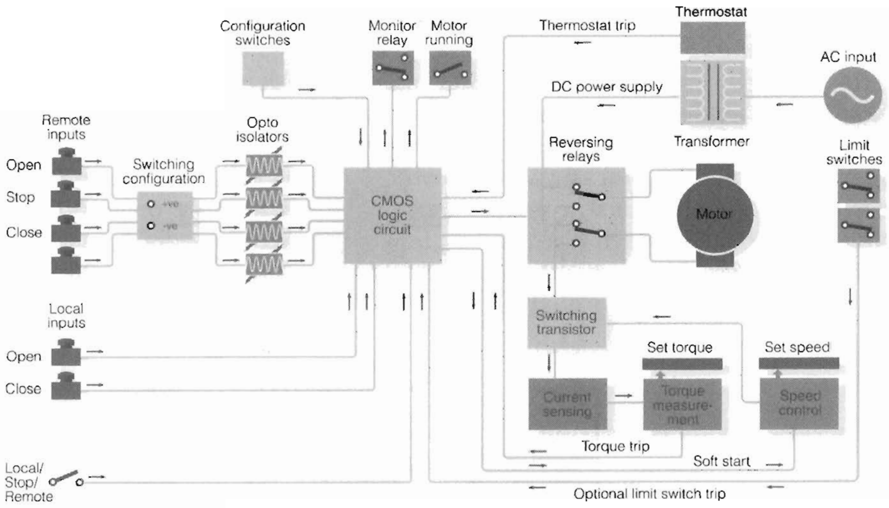


Figure 10. Circuitry of electric actuator shown in Figure 9.

Essentially, the non-intrusive actuator makes it unnecessary to remove electrical covers during motorised valve commissioning by providing local controls that do not penetrate the electrical enclosure.

The actuator consists of a self-contained unit for intermittent valve operation and comprises a three-phase motor, reduction gearing, reversing contactor starter with local controls and monitoring facilities, all based in a double-sealed, watertight enclosure to IP68 NEMA 4 and 6.

All torque and turn settings and configuration of the indication contacts are effected using a non-intrusive, handheld infra-red setting tool.

Infra-red setting

The infra-red programming and setting device is a handheld setting tool that allows the valve actuator to be configured, interrogated and commissioned in a completely non-intrusive way. This allows, for example, the possibility of making adjustments to a 'live' actuator within hazardous and wet areas (operational conditions permitting). A liquid crystal display on the actuator shows its status digitally. LEDs also show, for example: green—fully closed, yellow—any intermediate position and red—fully open. The handheld infra-red setting tool can confirm torque settings and perform simple diagnostic procedures to reveal why an actuator may not be functioning correctly. Power-supply faults, interlocks and other interruptions can be identified. Torque levels, limit settings and the configuration of the actuator may also be changed. Each time the actuator is powered up, it automatically tests its operational circuit's memory devices.

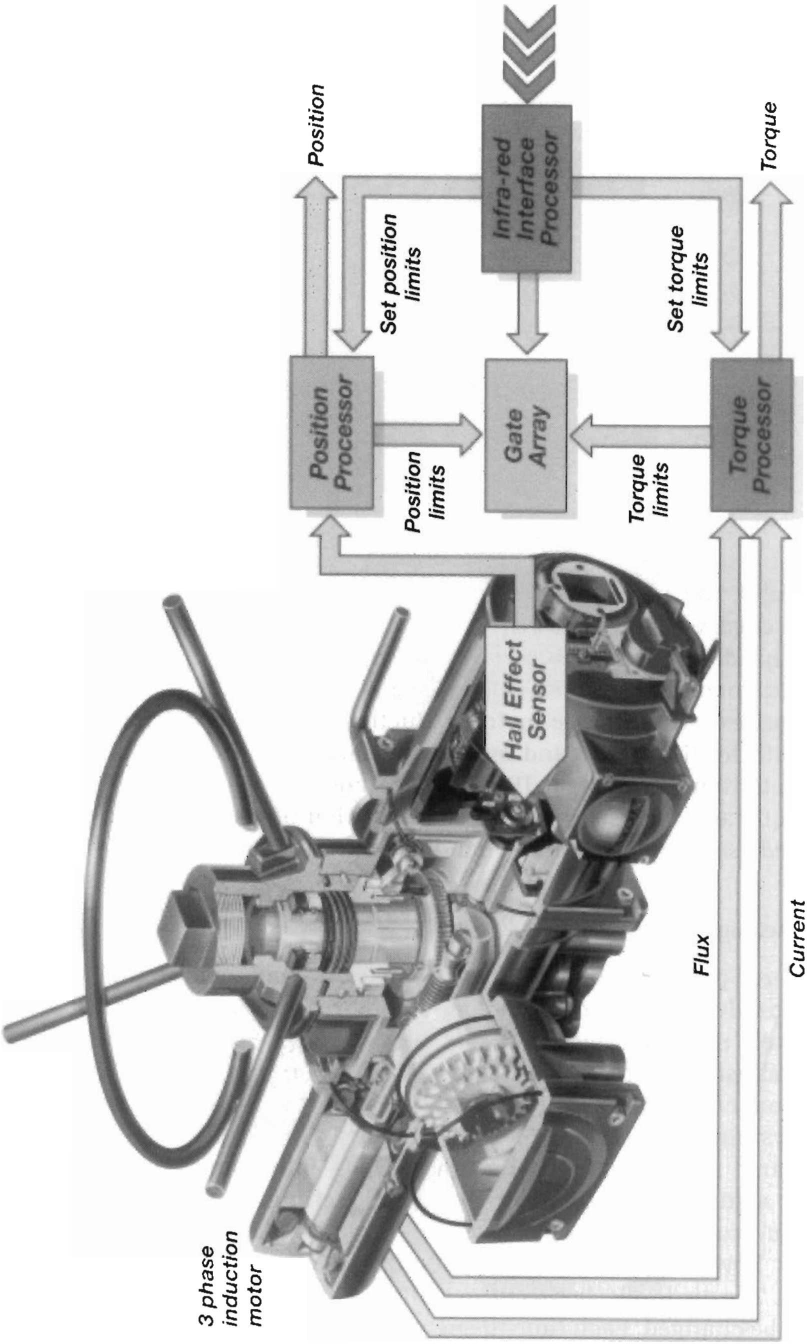


Figure 11. Electrically-operated, electronically-controlled intelligent actuator.

Handheld interrogation computers allow actuators to be assessed via an infra-red link which is connected to the handheld computer and attached to the actuator indication window (Figure 12).

This unit carries out all the functions previously described and downloads them for analysis. Historical information such as operator actions and output torque profiles can also be downloaded for analysis and to provide a view of operating events that can be viewed in groups or as traces. Details of a valve's last opening/closing cycle profile and its historical average opening/closing profile are also captured. From these, the latest profile could be compared with the average, giving an idea of changes in the valve torque requirement.

Communication and supervisory control

The major milestone in flow-control development and technology has been the introduction of fieldbus control systems.

Modern facilities require up-to-date communications right down to plant level.

Fieldbus systems take advantage of the latest developments in low-cost micro-chip technology, allowing intelligence to be built into each device in the loop.

This intelligence allows the rapid development of networks of intelligent control devices, each communicating with each other over a common medium.

Figure 12. Handheld computer gives access to valve-actuator diagnostics and configuration.

The 'peer-to-peer' control allows operators accurately to position a valve and thus achieve the optimum process parameters, including temperature levels, pressure and other critical variables.

Each control device communicates via a protocol that has been specially developed for the reliable and quick transmission of control information, with guaranteed transmission times for high-priority messages.

Fieldbus systems promise greater process control coupled with increased accuracy, efficiency, cost savings and plant reliability. The diagnostic role of fieldbus means that it can also give advance warning of potential problem areas, thus enabling preventative action to be taken.

Two-wire loop systems

Two-wire communication systems provide the link between valve actuator and supervisory control. Systems generally have three essential elements—field units, the 2-wire loop and a master station.

Field control units are typically mounted within the actuator's housing. Variable parameters such as the field unit address and baud rate are set non-intrusively using an infra-red communications link. Changes to the parameters can usually only be made when the field unit is in 'loopback' mode. An EEPROM holds the address and communication speed data, and a detector senses the loop current. The field unit does not interfere with the actuator local controls which remain operable in the event of field-unit malfunction.

The 2-wire loop carries a current loop signal which is modulated by a master station to send and receive data from field control units. The cable is a single twisted pair with an overall protective screen. The loop is capable of being installed in an electrically hostile environment where large surge currents can induce transient currents in the cables (i.e. lightning storms). The use of a 2-wire system greatly reduces the number of cable cores to transfer signals from the actuator to the control centre.

The two wires are connected to, and taken from, each field control unit in turn. They originate from and return to the master station to create a single twisted-pair, 2-wire loop.

As each device may be accessed from either direction, a redundant communications path is available. The integrity of the 2-wire cable is continuously checked whilst the system is running. Should communication fail, the master station ceases transmission and every field unit asserts its 'loopback' circuit. After a short period, the master station begins communication to each field unit in turn, extending the current loop until the fault location is revealed.

The master station is typically equipped with two processors, one to control the loop data and the other to handle the host communications, screen display and keypad. All messages passed over the network are under the control of the master station. A field unit may not transmit any data unless it receives a

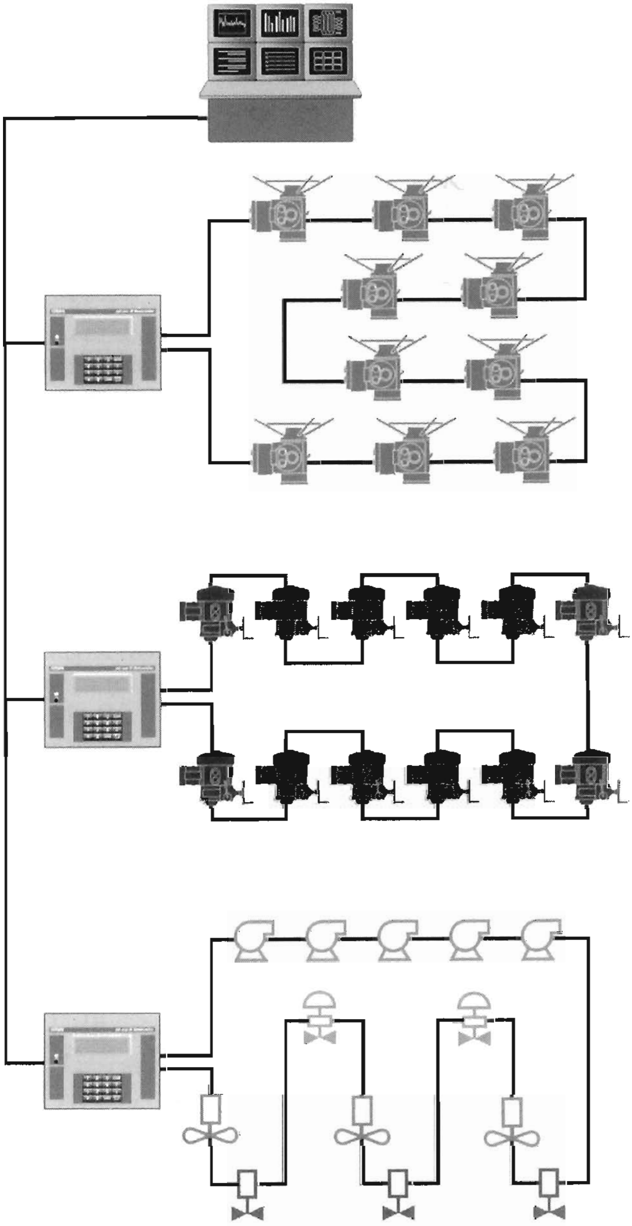


Figure 13. Advanced 2-wire loop-network communication system.

request from the master station. The host system may be a DCS, PLC or SCADA system. The information is typically passed using a universally accepted fieldbus communicator standard, e.g. Modbus, HART, INTERBUS, etc., protocol.

Information is continuously gathered by the master station from the field units, so ensuring that information requests by the host system are serviced with an intermediate reply from the internal data base.

Command instructions from the host have priority and are processed immediately by passing the message to the field unit concerned.

Advances in this technology also provide for high levels of system safety and security, including hot standby, cable fault protection and field unit failure protection, as well as logic and sequencing capabilities of a PLC, and Direct Operator Panels where operatives require push buttons and indicators for valve positions and graphical interfaces to the valves and plant using mimic diagrams to show the plant layout.

Any valve may be controlled and navigation through the application is by mouse control. Systems of this type can operate with a single network covering 240 devices over a 20 km loop length, without restriction on inter-node distances.

Some examples of layouts showing the control of actuators by Bus system are shown in Figures 14 and 15.

Electronic controllers have reached a high stage of development and sources within industry consider that the Fieldbus specification may be too complex in attempting an all-embracing standard. Problems are possible in migration between Fieldbus variants and integration with distributed control systems.

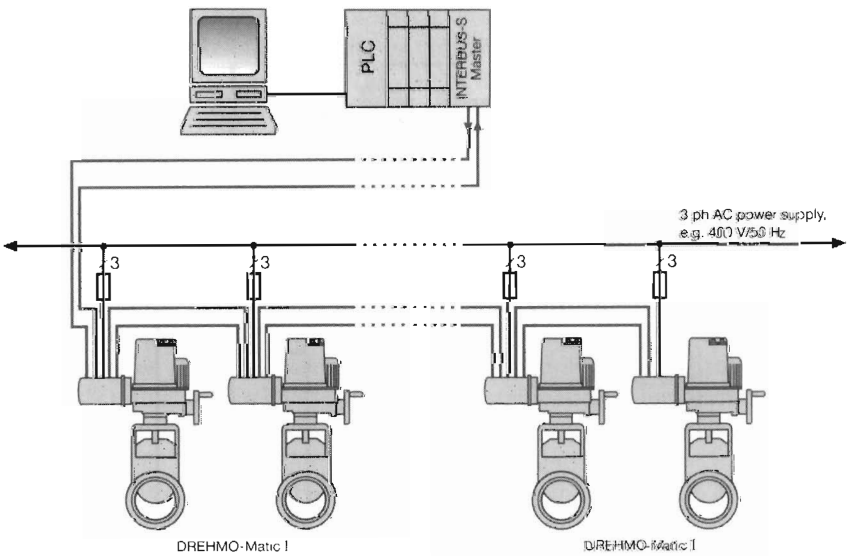


Figure 14. Controlling the actuators by BUS system: Interbus-S.

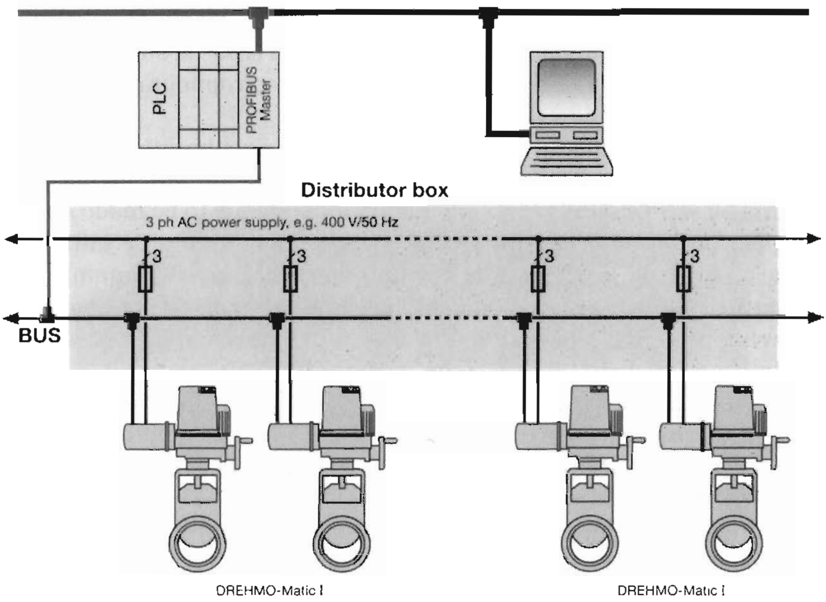


Figure 15. Controlling the actuators by BUS system: Profibus.

This is important where a transparent interface between the intelligent 'smart' valve and the Distributed Control System is required.

There is also concern that lightning strikes could cause failure of the microprocessors in the positioner. Manufacturers are actively working in this area developing their own intelligent electronics systems such as Fieldview, Starpak, Smart Valve Interface, Pakscan, ISMO, TZID, Keydig and Matic.

Valve positioning

There are two basic requirements in the opening and closing of valves: when closing, to be sure that the valve is properly and tightly seated, yet without excessive force being applied; when opening, to be certain that the valve is fully opened without excessive overrun or strain on the backseating.

A positioner is a device for varying and maintaining an actuator position in control valve applications. The positioner compares the actual actuator position with respect to the given input signal and adjusts the pressure applied on the actuator until the desired position is attained. Positioners can be pneumatic or electropneumatic, single- or double-acting and capable of being used on both rotary and linear actuators. Typically, a pneumatic positioner is a single-stage, force-balance device that can regulate virtually any actuator stepless from 0 to 100%. In basic form it consists of a flapper and nozzle, spool pilot valve, an adjustable range and reversing mechanical feedback. Generally, the electropneumatic positioner's action is based on the principle of analogue electronic comparison. More advanced electropneumatic positioners are low-powered high-flow switching devices.

Using menu-driven button control pads or remote communications devices, the operator selects from a wide range of pre-set or automated control characteristics. User configuration enables adjustments to be made manually so that parameters can match specific process conditions. Certain systems have the ability to adjust the valve's position using a static or dynamic option. Static setting allows the fixed adjustment of the actuator's upper and lower limits, while the dynamic option enables the valve position to be altered to match the individual installation requirements.

The valve positioner shown in Figure 16 is an intelligent digital device that combines micro-processor technology with a piezo-electric interface. The instrument can be connected to 2-, 3- or 4-wire systems and checks to establish what it is connected to and what is required of it, then calibrates the basic settings accordingly. Some positioners incorporate or can accommodate a gauge block directly onto the positioner for a continuous indication of the input and output air pressure of the actuator and the positioner.

Electric modulating actuator for control valves and damper applications.

Programmable positioner for quarter-turn actuators.

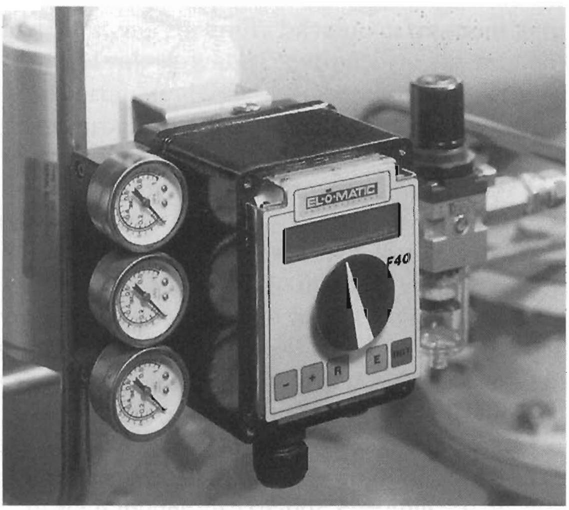


Figure 16. Electropneumatic valve positioner incorporating digital technology.

Limit switches

A limit switch is a device connected to an actuator or valve that transmits a signal when the valve reaches a pre-established position.

A typical limit-switch box for pneumatic actuators is fitted with two switches. These are activated by two adjustable cams mounted on the drive shaft. A direct coupling with the actuator drive shaft provides an exact indication of the valve's position.

The switches can normally be set independently of each other to provide an open/closed signal of the drive shaft and thus the position of the valve to the control room.

Gear operators

There are three main types of gear operators. These are of worm gear, bevel gear and spur gear design. Gear operators are generally suitable for both manual and motorised use (Figure 17).

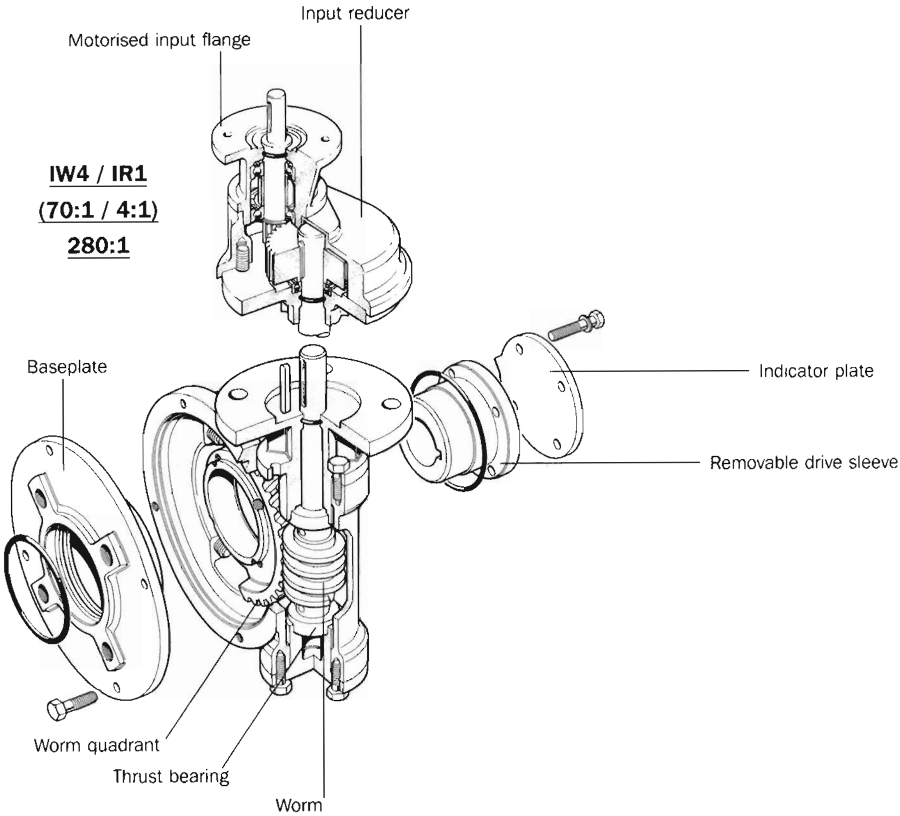


Figure 17. Quarter-turn worm-gear operator.

Worm-gear operators tend to be used with butterfly and ball valves and dampers, as well as other applications where keyed shafts are used to operate equipment.

Spur, bevel and multi-turn worm-gear operators are for use on gate, globe, sluice and penstock valves, as well as other applications where screwed or keyed shafts are used to operate equipment. Applications include low and high temperatures, submersible duties, buried service, marinised duty, water works specification and special indication.

Efficiency

It has been said already that some valve motor drives are mechanically inefficient—giving only 10% useful energy when, for instance, they are driving through a worm gear and the nut-and-screw of, say, a gate valve, and not much higher when driving a butterfly valve through self-locking gears. Wearing of inefficient gears or stem nuts is the primary limitation on frequency and continuity of operation of such drives, and this also limits the practical speed of operation of large screw-operated valves and penstocks. Nevertheless, the many advantages of control, power source, interfacing with supervisory control and instrumentation and so on compensate for the mechanical inefficiency, provided the actuator duty is properly considered.

Portable valve actuators

Portable valve actuators or valve wrenches can be used to provide portable turning power to replace manual effort in opening or closing valves on pipeline systems not fitted with permanent valve actuators. For example, water distribution piping systems generally have no permanent actuators installed, opening and closing of valves, when required, being by manpower.

An advantage of portable valve actuators is that they permit implementation of valve-exercising programmes designed to service and operate valves in order to keep them in good condition, e.g. eliminate 'valve seizure' problems which can arise when valves are left in one position for long periods. This applies particularly in water systems.

Portable valve actuators and valve wrenches may be designed for operation by electricity (i.e. powered by electric motors), compressed air or hydraulic power, and are normally adaptable for fitting a wide range of standard valves. The best designs provide operation 'feel', as well as speed control, reversibility and the ability to make instant stops and starts and reverses needed to free up sluggish gate valves, hydrants and sluice gates, etc. Typically, speeds of up to 20 to 25 r/min may be provided, with operating torques up to 1360 Nm (1000 lbf/ft) for handling valves in the size range 152 to 1524 mm (6 to 60 in).

The majority of such models are readily portable, i.e. weigh less than 18 to 23 kg (40 to 50 lb). Larger, heavier models may be trolley- or truck-mounted.

Control Valves

In general terms, a control valve may be described as a power-actuated valve, capable of throttling or modulating the flow, and used as a final control element in a control loop. Control valves operate automatically, receiving signals from an external controller. They may incorporate several different types of valve including globe, butterfly, ball and diaphragm (Figure 1).

Their operation can be vacuum, pneumatic, electromagnetic and hydraulic.

Within the area of control valves, most developments have taken place with the internal components such as the trim, which may well be of the cage type (hollow cylindrical trim), with retained seat, instead of contoured plugs with threaded seats.

Development in control valves has been driven by a demand for higher temperatures and pressures within the chemical, oil and power industries. Transcontinental pipelines, offshore oil and gas platforms and under-sea modules have been important driving forces.

Developments for control valves point to electronic packages located on valves which can be called up for remote interrogation. 'Smart valves' accessible by a

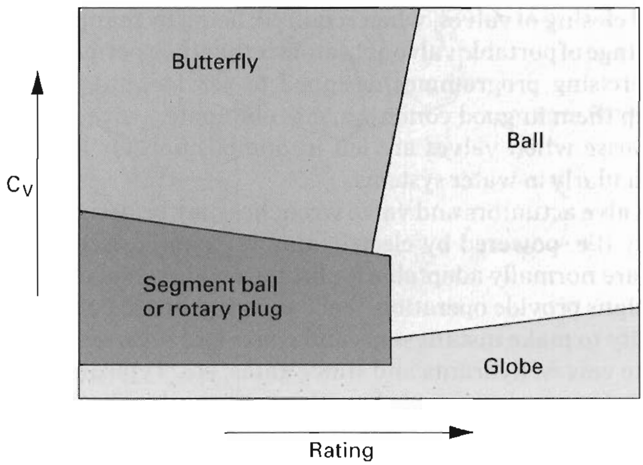
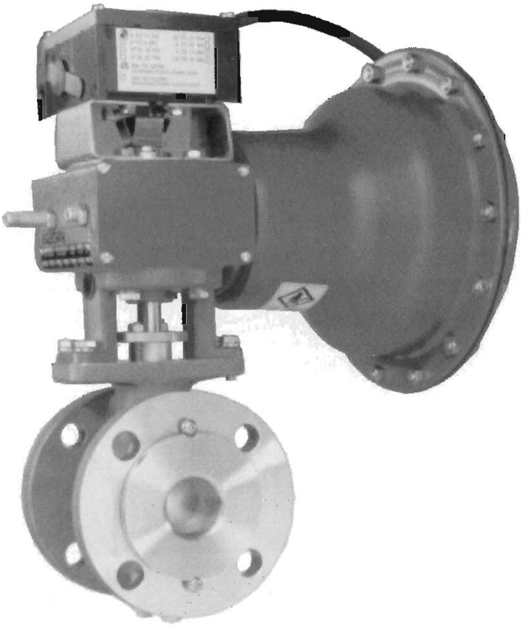
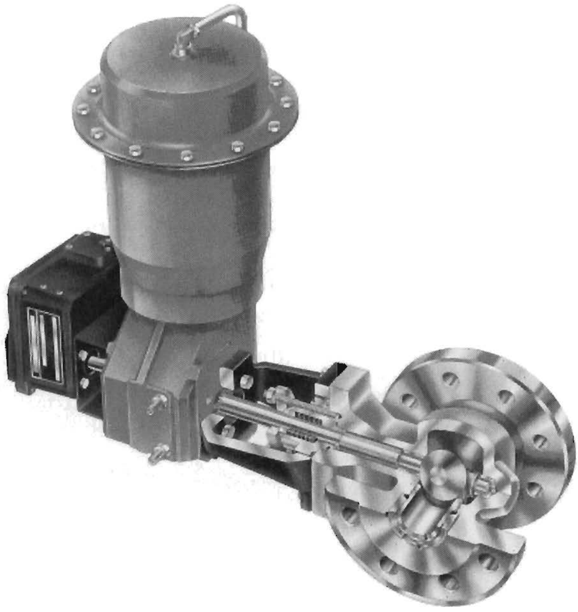


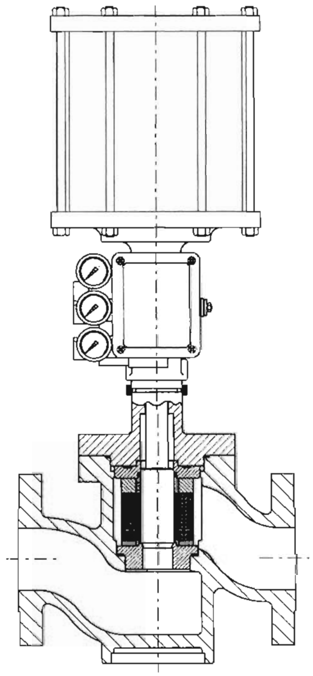
Figure 1. Control valve types.



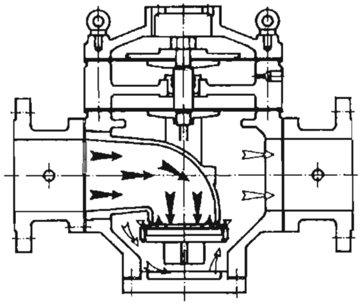
Control valve.



Rotary control valve.



Severe-service control valve for oil and gas duties.



Pressure control valve.

Flow control valve.

digital link receive and transmit commands that can indicate a stroke check of the valve, recording pressure vs valve travel as the valve is stroked. With an additional sensor for stem and shaft position, the data can be used to evaluate the condition of the actuator and accessories under various parameters based on an investigation of the valve's stored history. Digital communications to the valve can measure input signal, pneumatic pressure and valve travel, comparing the data with stored, expected values and recommending corrective action.

Current difficulties arise over the fieldbus communications standard. While the concept of valve intelligence does not depend on digital communications, the reality is that some form of digital fieldbus is essential.

Control valve sizing

A number of valve manufacturers have produced control valve sizing programs that can be used to select a control valve with optimum controllability and control accuracy for each process application. Typically, these programs are based on the flow characteristic curve and gain curve of the installed valve.

Inherent and installed flow characteristics

The selection of a control valve of optimum size and type begins with the valve's flow characteristic. This has been defined as the curve relating percentage of flow to percentage of valve travel, i.e. rotation of the ball or the butterfly disc or linear movement of the globe valve disc.

'Inherent flow characteristic' applies to situations when constant pressure drop is maintained across the valve.

'Installed flow characteristic' takes into account the variations in the pressure drop caused by conditions in the system where the valve is installed.

There are two common flow characteristics for control valves. In the equal percentage characteristic, a given fraction of valve opening changes flow by a certain percentage of previous flow. In the linear characteristic, a given fraction of opening changes flow by the same fraction of maximum flow.

Figure 2 shows the most common valve inherent flow characteristics as a function of the relative flow coefficient (ϕ) and the relative travel (h).

A typical installed flow characteristic curve for a butterfly valve in a process application is illustrated in Figure 3.

Overshoot

An important function of control valve performance is 'overshoot'.

As the control valve responds to a step change in signal, overshoot is the amount of travel beyond the final steady-state position. While in most systems it is important that a control valve responds quickly to changes in signal, it is equally important that this response does not destabilise the operation.

Excessive overshoot can contribute to loop nonlinearity, as well as increase loop instability and affect control-loop performance.

With a signal of 5%, a control valve with zero overshoot will travel directly to the required position. A valve with up to 20% overshoot will pass beyond the proper position by 1% and require time for adjustment to the new position. A control valve with the least amount of overshoot will provide the system with the best opportunity to respond to changes in process demands.

Percentage overshoot should be less than or equal to 20% of the step magnitude for steps ranging from 10% of travel down to steps equal to the backlash/stiction limit +1%. Speed of response needs to be viewed in conjunction with how accurately the control valve responds to a change in input signal and the percentage of valve overshoot. Quick response by itself without a high level of accuracy and with too much overshoot will destabilise system performance. Control valve speed of response is typically determined by four major criteria:

- Dead time (Td)—the time it takes to respond after the signal is initiated. It is measured in fractions of a second.
- T63—the time it takes the valve to reach 63% of its new position. It is measured in seconds or fractions of a second.

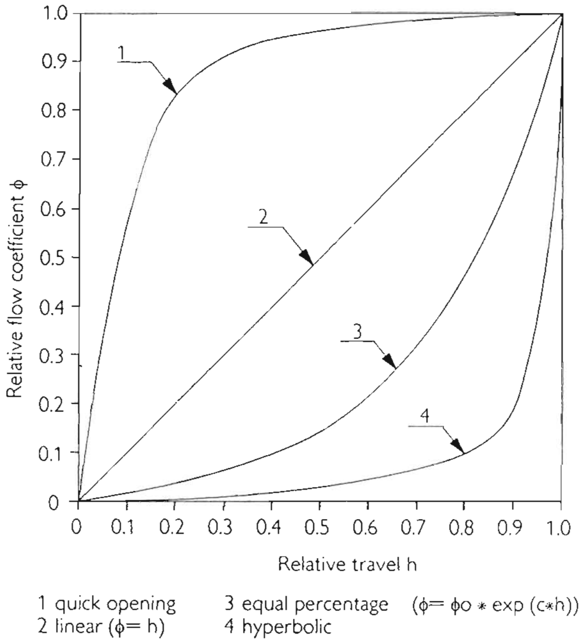


Figure 2. Valve inherent flow characteristics.

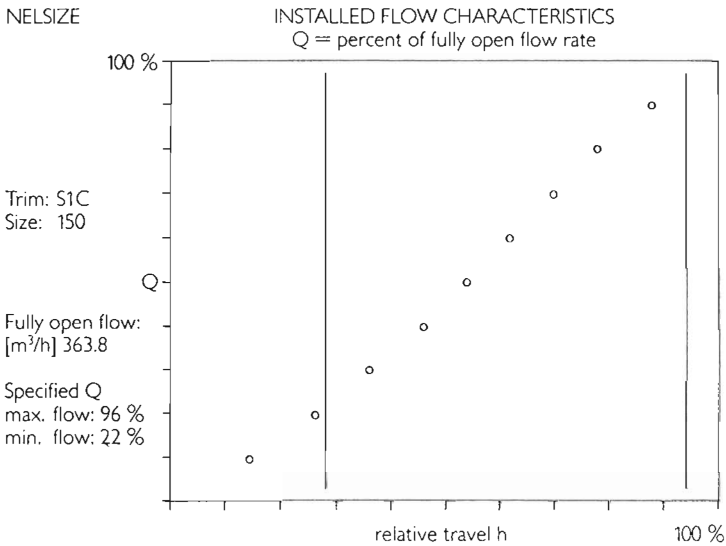


Figure 3. Typical installed flow characteristic for a butterfly-control valve.

- T98—the time it takes the valve to reach 98% of its new position. It is measured in seconds or fractions of a second.
- Band width—the frequency at which the plug position amplitude is diminished by 6 dB (difficult to verify).

The control valve shown in Figure 4 has been designed for highly erosive services and with simple maintenance procedures without special tools. The valve can be used with sodium chlorate, wet chlorine, terepthalic and hot acetic acid because it has a tiodyed titanium seat and plug, a ceramic-coated titanium shaft and titanium bearings with Kalrez®* seals.

The plug and seat are self-aligning and bi-directional flow capability helps to prolong valve life. The valve operates over a 90° range of motion and can

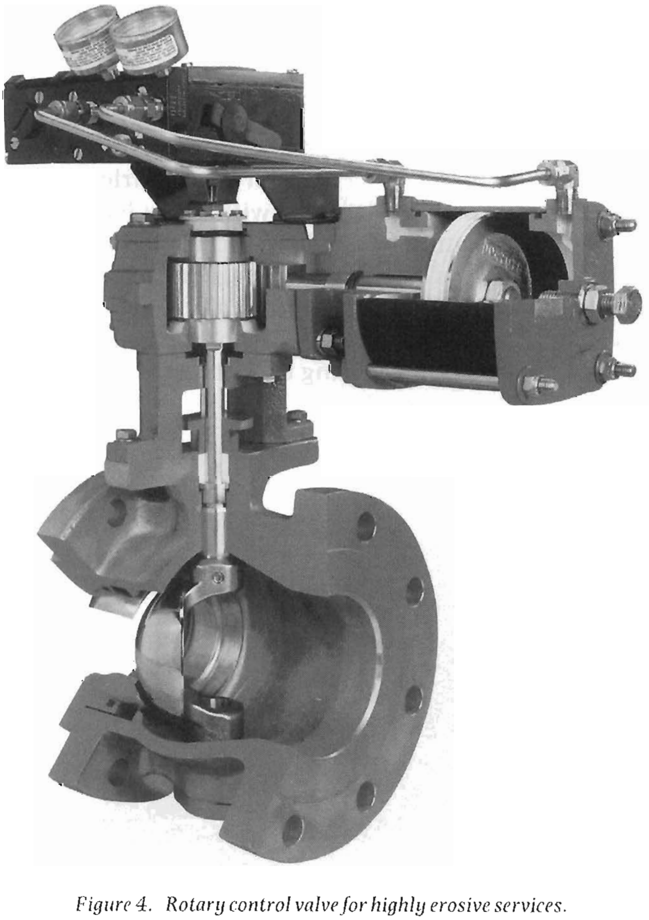


Figure 4. Rotary control valve for highly erosive services.

*Dupont registered trade mark.

withstand high pressure drops. Typical products being handled include digester gas off, Kaolin slurry, fly-ash slurry, titanium dioxide slurry and steam.

Segment control valve

This type of valve may be considered to be a special category of single-seated control valve (Figure 5). Segment valves have a concentric V-shaped trim on the side of a rotary ball segment (Figure 6). They are typically designed as high-capacity, general-purpose valves with equal percentage flow characteristic. An almost linear characteristic can be obtained using the non-V-shaped cheek of the ball.

The metal-seated butterfly control valve shown in Figure 7 is equipped with a flow-balancing trim. The trim is located on the downstream side of the valve body and is effectively a perforated plate partially covering the valve outlet to reduce noise and cavitation, stabilise the flow pattern and reduce dynamic torque on the disc. Valve sizes range from DN 150 to DN 100 (6 in to 40 in).

The concept of this design is to transfer fluid forces out of the disc to the body. Figures 8 and 9 illustrate flow treatments with a concentric-type, conventional butterfly valve compared with a valve fitted with a flow-balancing trim.

Figure 10 shows the seating principle of this butterfly control valve. The disc of the valve is machined to close tolerances to create an elliptical shape similar to an oblique slice taken from a solid metal cone.

When the valve is closed, the elliptical disc at the major axis displaces the seat ring outwards, causing the seat ring to contact the disc at the minor axis.

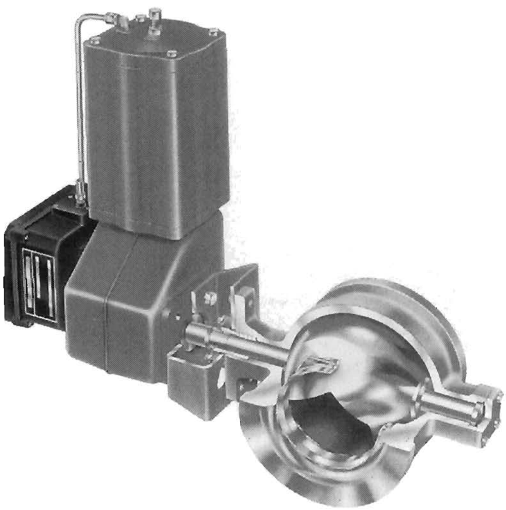


Figure 5. Segmented control valve.

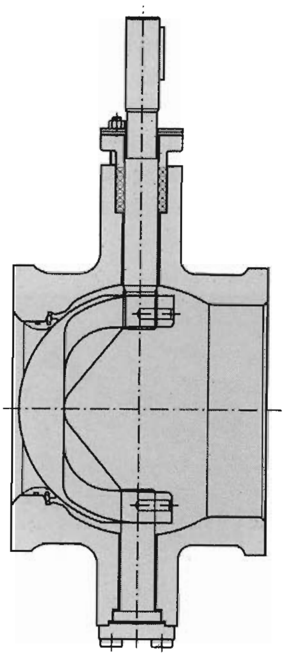


Figure 6. Concentric 'V'-shaped trim on side of rotary ball segment.

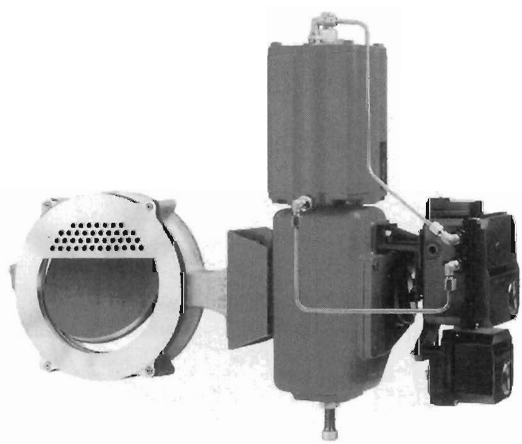


Figure 7. Metal-seated butterfly-control valve with flow-balancing trim.

When the valve is opened, the contact is released and the seat ring returns to its original circular shape.

Double-seat valves

This type of valve (Figure 11) employs two seats with an optimally dimensioned leakage chamber between them so that any seal leakage drains directly to atmosphere.

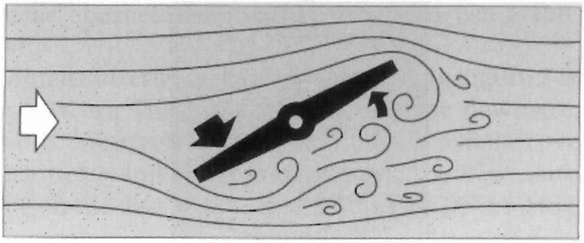


Figure 8. Conventional concentric-type valve flow treatment.

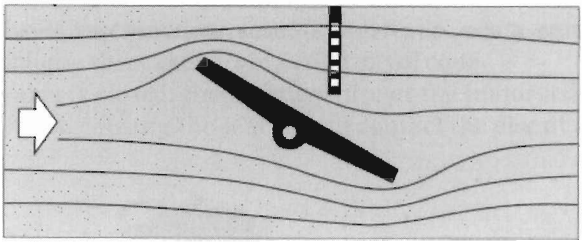


Figure 9. S-DISC flow treatment.

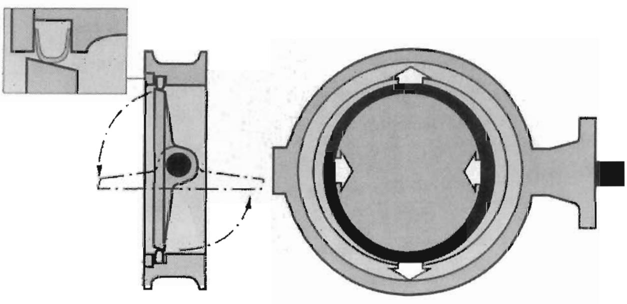


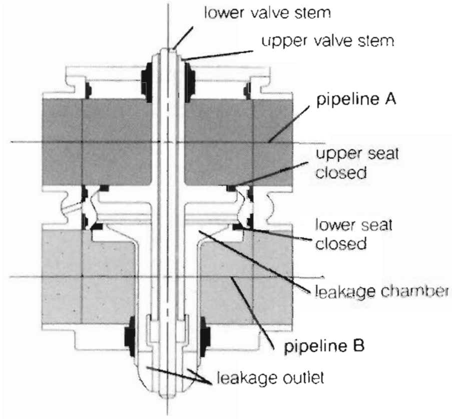
Figure 10. Seating principle.

The valve is pneumatically operated and suited for applications in brewing, beverage, dairy, food, pharmaceutical and chemical plants.

The method of operation is shown in Figure 12. The valve is designed for cleaning in place (CIP), as outlined in Figure 13. Double-seat valves typically allow for the design of 'totally contained flow systems', eliminating the need for manual swing bends for product and cleaning lines, thereby simplifying the processes of flow automation.

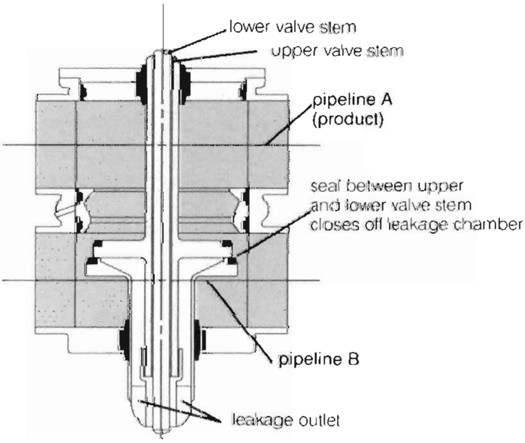
Figure 11. Double-seat-type control valve.

Double Seat Valve Operation



VALVE IN CLOSED POSITION.

Upper and lower valve seats are closed and separated by the leakage chamber open to atmosphere
In this position, two fluid streams (pipeline A and pipeline B) may pass through the valve without a chance of intermixing.

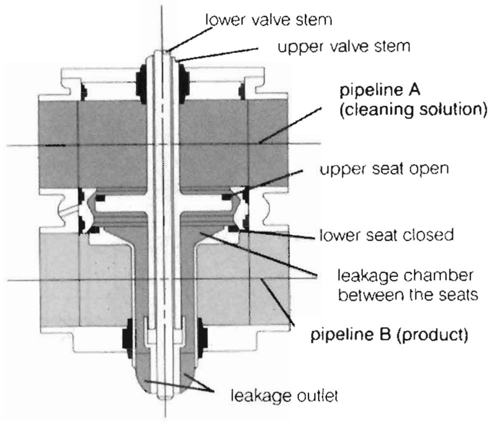


VALVE IN OPEN POSITION.

Upper valve stem moves down towards the lower one and presses it from its seat. At the same time its seal closes off the leakage chamber and separates it safely from the product area. In this position, fluid streams (pipeline A and pipeline B) are open to each other.

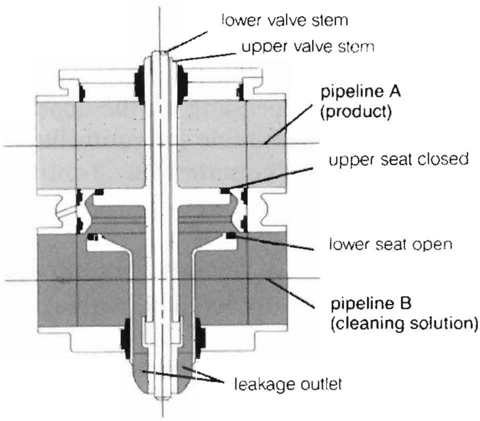
Figure 12.

Double Seat Valve Cleaning



UPPER VALVE SEAT AREA.

The upper valve stem is momentarily opened by the seat lift actuator. Cleaning solution is allowed to flush the upper seal and seat area as well as the leakage chamber.



LOWER VALVE SEAT AREA.

The lower valve stem is momentarily opened by the seat lift actuator. This allows cleaning solution to flush the lower seal and seat area as well as the leakage chamber.

Figure 13.

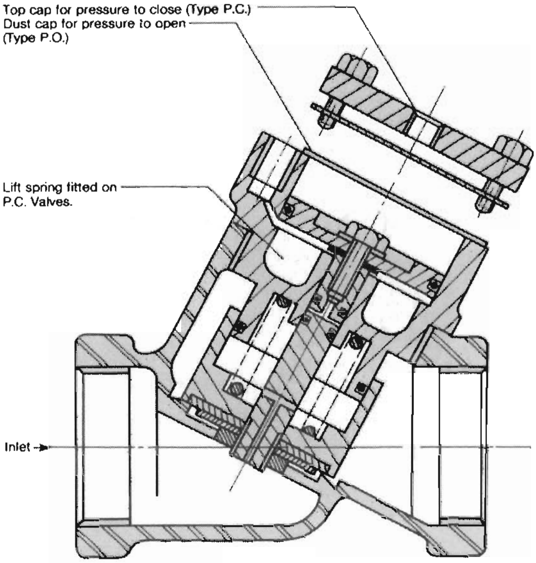


Figure 14. Piston-operated control valve.

Piston-operated control valves

The piston-operated control valve shown in Figure 14 is another valve developed to meet the needs of the processing industries. Power is applied through integral piston operators and the operating source can be pneumatic, hydraulic or line pressure itself, depending on the application. The valve is non-concussive in operation and suitable for controlling water, gases and other liquids compatible with the valve materials.

Modulating control valves

Control valves of this type are generally available with a comprehensive range of options.

They have a vertical pneumatic piston-type actuator with integral electronic controller/positioner housed in the head. The control head consists of an electronic processor that enables the valve to operate in either a 'positioner' or 'controller' mode (Figure 15).

Curve characteristics and signal scaling are set by means of dip switches.

The electronic controller's built-in software generates an adoptive response to changing process conditions, reducing hunting and overrunning.

The pneumatic-diaphragm type provides conventional mechanical control.

Control valves are used for a multitude of applications, e.g.:

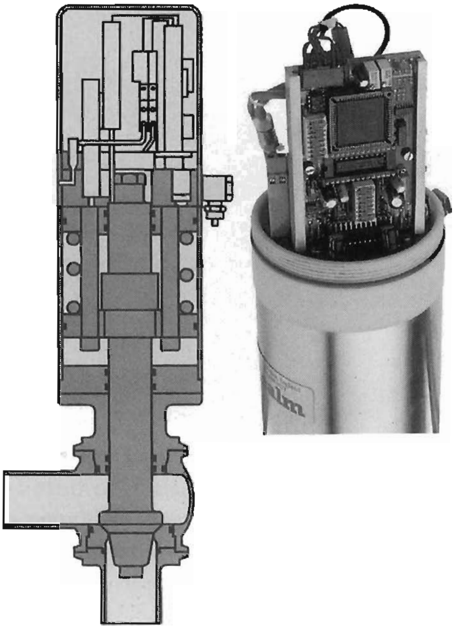
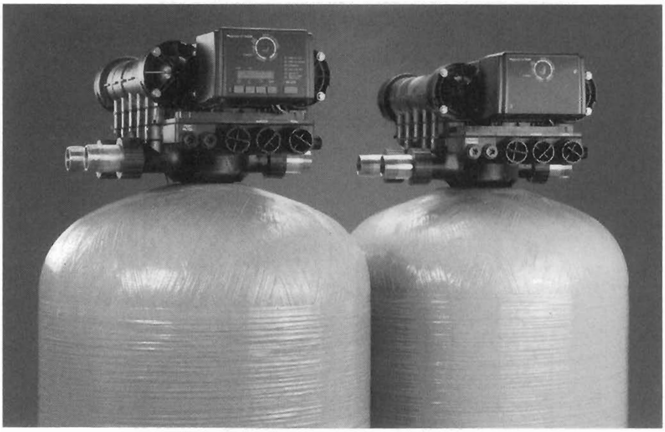


Figure 15. Modulating electronic control valve.

- Pressure control, including
  - Controlling
  - Downstream reducing and stabilising
  - Upstream sustaining
  - Holding a differential pressure
  - Backflow prevention
  - Double direction flow if upstream pressure < downstream pressure.
  - Full opening at a present upstream pressure

- Flow and level control, including
  - Controlling
  - Maintaining a maximum flow
  - Reducing and stabilising flow downstream
  - Controlling the upper level
  - Backflow prevention

- Tank and reservoir control, including
  - Controlling
  - Not controlling (fully open or fully closed)
  - Controlling the upper level
  - Opens at lower level, closes at upper level



Diaphragm control valves for softening and filter applications.

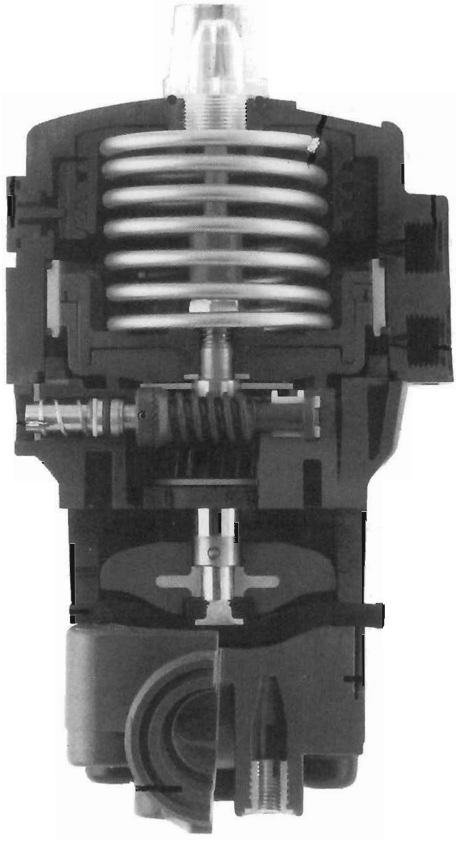


Figure 16. Actuated plastic-diaphragm control valve.

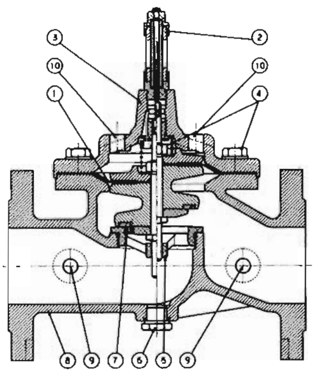
- Protection and control, including
  - Against water hammer
  - Against electrical failure
  - Pump protection
  - Electrical/electronic monitoring
  - Against downstream pipe failure
  - Against 'overspeed flow'

Pressure-reducing valves and pressure-relief valves are also used as control valves and are covered in their specific chapters. Trim designs include metal-seat ring and cage, soft seat and cage, balanced and unbalanced plug, soft-seated plug and bonnet types including bellows-seat bonnet.

The valve shown in Figure 16 is an actuated plastic-diaphragm control valve. Utilising a threaded spindle which operates as a helical gear, it is possible to regulate the minimum flow (fully closed or fully open). The stroke limiter in the upper part of the actuator is also adjustable. The internal gear operates as stroke limiter which provides for restriction as it rotates and thus provides stroke limitation from fully open to fully closed across its complete movement.

Typical applications include water treatment, water supply, and the chemical industry.

Iris-type control valves are used where it is difficult to control and regulate viscous or charged liquids as well as gaseous media, particularly if very small



1. Membrane: reinforced nitrile.
2. Position indicator with purge: brass and stainless steel.
3. High-pressure valve head (pressure setting 25): cast iron epoxy coating.
4. Nuts and bolts: stainless steel.
5. Replaceable streamlined seat: bronze.
6. Body drain plug: brass.
7. Reversible seat seal: nitrile.
8. High-pressure valve housing: cast iron epoxy coating (pressure setting 25).
9. Pressure relief holes.
10. Pressure relief holes.

Water industry control valve.

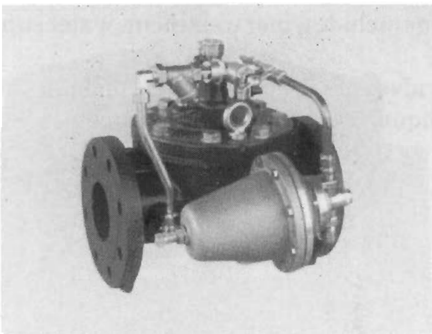
quantities are involved. Conventional flow control valves with slot-type, sickle-shaped or elongated flow apertures will not generally permit fluctuation-free regulation. Other control valves tend to have greater head losses that can result in increased energy costs.

A typical iris-design control valve is shown in Figure 17.

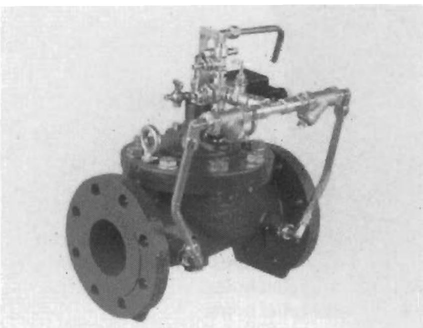
It will be seen from Figure 18 that the flow aperture is almost continuously variable, making it suitable for use in sugar centrifugals, sewage treatment stations, paper and paper board processes, etc.

Pneumatic control valves

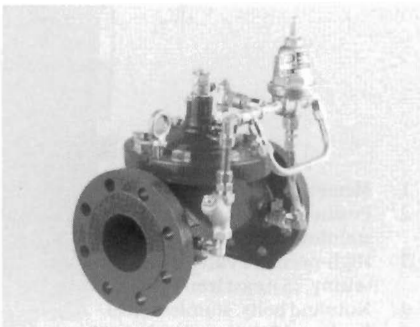
Pressure-operated and solenoid air poppet and spool valves are extensively covered in the 'Pneumatic Handbook', also published by Elsevier Science Limited, which should be referred to for information on this subject. The valves covered include 2- and 3-port, direct and pilot-operated and pressure-operated valves as well as 4- and 5-port solenoid air-operated valves.



Progressive opening and closing altitude valve.



Pump protection control valve.



Upstream/downstream control valve.

Examples of water industry standard control valves.